CHAPTER 8

ELECTROMAGNETIC WAVES

CHAPTER ESSENCE AT A GLANCE:
Maxwell's displacement current, transverse electromagnetic waves, wave velocity relations, and the spectrum from radio waves to gamma rays.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Displacement Current & Maxwell's Extension:** Changing electric fields produce magnetic fields just as physical moving charges do, maintaining current continuity across capacitor dielectric gaps.

• **Transverse Nature of Electromagnetic Waves:** Self-propagating electromagnetic waves consist of mutually perpendicular electric and magnetic field oscillations that are both perpendicular to wave travel.

• **Electromagnetic Spectrum Continuum:** Continuous frequency spectrum propagating at the speed of light in vacuum, spanning radio waves, microwaves, infrared, visible, ultraviolet, X-rays, and gamma rays.

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = 3 \times 10^8 \text{ m/s}$$

Speed of electromagnetic waves in vacuum

$$c = \frac{E_0}{B_0}$$

Ratio of electric to magnetic field amplitudes in EM wave

$$I_d = \epsilon_0 \frac{d\Phi_E}{dt}$$

Maxwell's displacement current

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Inside the dielectric gap of a charging capacitor, conduction current I c = 0, but displacement current I d = I c.

• **Alert:** [Valuation Rubric] Identify radiation correctly by wavelength or frequency range .

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Concept of displacement current and Maxwell's modification of Ampere's law [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Inconsistency of Ampere's Law]** Ampere's circuital law states closed integral B · dl = μ_0 I c. Maxwell noticed that during charging of a capacitor, a loop around the wire gives B ≠ 0, but a surface bulging between capacitor plates has no conduction current (I c = 0 implies B = 0), creating a logical contradiction. [1 Mark]

• **[Displacement Current Concept & Formula]** Changing electric field between plates creates changing electric flux Φ E. Φ E = E A = (Q divided by ε_0 A) A = Q divided by ε_0 implies Q = ε_0 Φ E. Rate of change of charge: I d = dQ divided by dt = ε_0 (dΦ E divided by dt). This current arising from time-varying electric field is called Displacement Current (I d). [1 Mark]

• **[Ampere-Maxwell Law]** closed integral B · dl = μ_0 (I c + I d) = μ_0 [I c + ε_0 (dΦ E divided by dt)]. Outside plates: I = I c, I d = 0. Between plates: I = I d, I c = 0. Total current is continuous! [1 Mark]

• **Key Exam Result:** I d = ε_0 (dΦ E divided by dt); Ampere-Maxwell Law: closed integral B · dl = μ_0(I c + I d)

2. Maxwell's equations (integral forms and significance) [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Four Maxwell's Equations in Integral Form]** 1. Gauss's Law in Electrostatics: closed integral E · dA = q in divided by ε_0. (Electric charges produce electric fields; isolated charges exist). [1 Mark]

• 2. Gauss's Law in Magnetism: closed integral B · dA = 0. (Magnetic monopoles do not exist; magnetic field lines form closed loops). [0.5 Mark]

• 3. Faraday-Maxwell Law: closed integral E · dl = - dΦ B divided by dt. (Time-varying magnetic field induces an electric field). [0.5 Mark]

• 4. Ampere-Maxwell Law: closed integral B · dl = μ_0 I c + μ_0 ε_0 (dΦ E divided by dt). (Magnetic fields are produced by conduction currents AND time-varying electric fields). [1 Mark]

• **Key Exam Result:** Maxwell's 4 equations unify Electricity, Magnetism, and Optics

3. Transverse nature of EM waves and relation c = 1 divided by √(\μ 0 varepsilon 0) [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Transverse Nature of EM Waves]** In an electromagnetic wave propagating along the x-axis, the electric field oscillates along the y-axis (E = E y unit vector j) and the magnetic field oscillates along the z-axis (B = B z unit vector k). Both E and B are perpendicular to each other and mutually perpendicular to the direction of propagation (unit vector i = unit vector j × unit vector k). This proves the wave is transverse in nature. [1 Mark]

• **[Speed in Vacuum]** Speed of EM wave in vacuum is determined solely by electric and magnetic constants: c = 1 divided by √(μ_0 ε_0) = 1 divided by √[$(4\pi \times 10^{-7})(8.854 \times 10^{-12})$] = 3.0×10^8 m/s. In a dielectric medium: v = 1 divided by √(μ ε) = c divided by √(μ r ε r) = c divided by n (where n is refractive index). [1 Mark]

• **Key Exam Result:** E ⊥ B ⊥ propagation vector; Speed in vacuum c = 1 divided by √(μ_0 ε_0) = 3×10^8 m/s

4. Relation between peak amplitudes of electric and magnetic fields E 0 = c B 0 [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Relation between Field Amplitudes]** The ratio of peak amplitude of electric field E_0 to peak amplitude of magnetic field B_0 at any point in an EM wave in vacuum equals the speed of light: E_0 divided by B_0 = c = 3.0×10^8 m/s ⇒ E_0 = c B_0. Instantaneous values are also locked in phase: E(x, t) = c B(x, t). [1 Mark]

• **[Numerical Example]** If $B_0 = 2.0 \times 10^{-7}$ T: E_0 = c B_0 = $(3.0 \times 10^8$ m/s) × $(2.0 \times 10^{-7}$ T) = 60 V/m (or $\frac{N}{C}$). Because c is enormous, the electric field amplitude is numerically much larger than magnetic field amplitude in SI units. [1 Mark]

• **Key Exam Result:** E_0 = c B_0; For $B_0 = 2.0 \times 10^{-7}$ T, E_0 = 60 V/m

5. Relation between peak amplitudes of electric and magnetic fields E 0 = c B 0 [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Relation between Field Amplitudes]** The ratio of peak amplitude of electric field E_0 to peak amplitude of magnetic field B_0 at any point in an EM wave in vacuum equals the speed of light: E_0 divided by B_0 = c = 3.0×10^8 m/s ⇒ E_0 = c B_0. Instantaneous values are also locked in phase: E(x, t) = c B(x, t). [1 Mark]

• **[Numerical Example]** If $B_0 = 2.0 \times 10^{-7}$ T: E_0 = c B_0 = (3.0×10^8 m/s) × (2.0×10^{-7} T) = 60 V/m (or $\frac{N}{C}$). Because c is enormous, the electric field amplitude is numerically much larger than magnetic field amplitude in SI units. [1 Mark]

• **Key Exam Result:** E_0 = c B_0; For $B_0 = 2.0 \times 10^{-7}$ T, E_0 = 60 V/m

6. Electromagnetic spectrum in order of increasing frequency or decreasing wavelength [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[EM Spectrum in Order of Increasing Frequency]** 1. Radio Waves (λ > 0.1 m, ν < 3×10^9 Hz). 2. Microwaves (λ: 0.1 m to 1 mm, ν: 3×10^9 to 3×10^{11} Hz). 3. Infrared Rays (λ: 1 mm to 700 nm, ν: 3×10^{11} to 4×10^{14} Hz). 4. Visible Light (λ: 700 nm to 400 nm, ν: 4×10^{14} to 7.5×10^{14} Hz). 5. Ultraviolet Rays (λ: 400 nm to 1 nm, ν: 7.5×10^{14} to 3×10^{16} Hz). 6. X-rays (λ: 1 nm to 10^{-3} nm, ν: 3×10^{16} to 3×10^{19} Hz). 7. Gamma Rays (λ < 10^{-3} nm, ν > 3×10^{19} Hz). [2 Marks]

• **Key Exam Result:** Radio --> Micro --> Infrared --> Visible --> UV --> X-ray --> Gamma (Frequency increases)

7. Numerical on frequency and wavelength of EM wave c = \ν \λ [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Formula & Problem]** Speed of light c = ν λ ⇒ λ = c divided by ν. Given radio transmitter frequency ν = 100 MHz = 1.0×10^8 Hz, c = 3.0×10^8 m/s. [1 Mark]

• **[Step 2: Substitution & Wave Region]** Wavelength λ = (3.0×10^8 m/s) divided by (1.0×10^8 s^{-1}) = 3.0 m. Since λ = 3.0 m is in the meter range, this radiation belongs to the VHF FM Radio wave band. [1 Mark]

• **Key Exam Result:** Wavelength λ = 3.0 m; Belongs to Radio wave spectrum